PREP 3 · LESSONS 5–6

Wires, series and parallel

Resistance of a wire (required practical) · rules for series and parallel circuits

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 State the standard units of potential difference and resistance. [1]

2 Describe how current depends on potential difference for an ohmic conductor. [1]

3 Describe the current at different points in a single closed loop. [1]

Key ideas LESSONS 5–6

Show your working

4 A student measures the resistance of a wire at different lengths.

Length / cm	20	40	60	80
Resistance / Ω	0.8	1.6	2.4	3.2

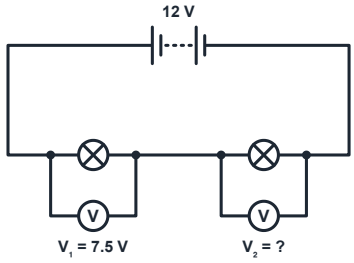
a Describe the relationship between resistance and length. [1]

b Name one control variable. [1]

c Explain why the potential difference is kept small. [1]

5 Two lamps are connected in series to a 12 V battery.

a Calculate the reading on V_2. [1]



b Two resistors, 4 Ω and 8 Ω, are in series. Total resistance: _____ [1]

c Describe the potential difference across components in parallel. [1]

d In a parallel circuit, the total current is 0.90 A. One branch has 0.40 A. The other branch has _____ [1]